Propedeutical Sandbox

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Abstract

In the digital ecosystem of Reply Mirror, the public health infrastructure MirrorLife represents a benchmark for data-driven prevention and sustainable well-being. Advanced cloud platforms integrate multimodal information streams — including lifestyle indicators, environmental exposures and longitudinal health signals — to support population-level health optimization.

In this interconnected environment, prevention is no longer episodic but continuous: health systems aim to anticipate deviations from optimal trajectories and promote timely, personalized interventions.

As part of The Eye initiative, you are called to design an intelligent system capable of monitoring dynamic well-being patterns, identifying emerging vulnerabilities and supporting proactive preventive strategies.

The objective is not to react to disease, but to optimize well-being trajectories across time through adaptive, evidence-based intelligence.

1 Problem Statement

It is 2087 and the digital metropolis of Reply Mirror operates under an integrated prevention framework. Institutions such as MirrorLife coordinate large-scale, longitudinal datasets to promote well-being optimization across the population.

The system continuously collects and harmonizes data related to daily routines, physical activity, sleep patterns, stress proxies, environmental context and social determinants. Within this complex and evolving landscape, subtle deviations from optimal behavioural or physiological trajectories may signal the need for preventive action.

You — The Eye — are tasked with developing an adaptive intelligence capable of identifying these early signals. The goal is to determine when a citizen's current trajectory suggests that proactive preventive support could improve long-term well-being outcomes.

Preventive risk patterns are not static. They evolve due to:

- changes in lifestyle behaviours and social habits;
- seasonal and environmental variations;
- demographic shifts and contextual factors;
- non-linear interactions between multiple low-intensity signals.

A key challenge lies in the temporal evolution of these patterns. Systems trained on historical data must remain robust when confronted with distributional shifts, behavioural drifts and structural changes in the environment.

Static predictive approaches are insufficient. The task requires adaptive, continuously learning systems capable of generalizing across time while maintaining calibrated and stable performance.

Challenge Flow The challenge unfolds in three levels. At each level, your team receives a training dataset representing the general well-being status of the citizens of ReplyMirror. In addition to this, you are provided with additional datasets containing varying levels of information about citizens.

In addition to the training data, each level also includes a corresponding evaluation dataset of comparable difficulty. These evaluation datasets are the only ones used to compute the official score for the level. For each evaluation dataset, only the first submission will be accepted and will be considered final.

Throughout all levels, you are asked to design a system of cooperative intelligent agents capable of classifying each individual-time instance according to the following outcome:

- 0 = continuation of standard monitoring (well-being trajectory within acceptable optimization range);
- 1 = recommendation for activation of a personalized preventive support pathway.

At the end of all three levels, the overall leader-board will reflect the resilience and adaptability of the system of each team.

Challenge Goal You are asked to design an AI-based system capable of:

- identifying dynamic patterns associated with suboptimal well-being trajectories;
- anticipating emerging preventive needs through longitudinal pattern recognition;
- adapting to structural and temporal shifts without performance degradation;
- maintaining a low rate of unnecessary preventive pathway activations.

Decision errors carry asymmetric implications:

- unnecessary activation of preventive support (false positive) may lead to inefficient allocation of resources and reduced system credibility;
- failure to activate support when appropriate (false negative) may result in missed opportunities for early optimization and long-term health benefits.

The final score combines predictive performance, temporal robustness and adaptability to evolving well-being dynamics.

2 Input Format

The input consists of a series of data set with increasing complexity. You have access to multiple types of data set, available for download in different formats. In detail:

Status.csv with T records of well-being monitoring events, including:

- Event ID: a unique identifier of the monitoring event;
- Citizen ID: a unique identifier of the monitored individual;
- Event Type:
  - routine check-up
  - preventive screening
  - lifestyle coaching session
- Physical Activity Index (aggregated daily score);
- Sleep Quality Index (aggregated score);
- Environmental Exposure Level (area-based indicator);
- Timestamp of the recorded event;

Locations with geo-referenced data, whereby the citizen has been located through GPS-based systems, irrespective of the type of activity performed, including:

- BioTag: a unique identifier of the citizen;
- Datetime: date and time of geo-referentiation;
- Lat: latitude coordinate in degrees;
- Lng: longitude coordinate in degrees;

Users with a resume of citizen's personal data.

3 Output Format

The output must be an ASCII text file. Each line should be separated by newline character and refers to Citizen ID with AI-recommended preventive support.

The format of each line is:

$$t$$

where:

- t represents the Citizen ID among the T record received in input.

4 Scoring Rules

The evaluation framework is built around a composite scoring model that measures performance across two key dimensions, with economic impact serving as the primary driver. Your goal is not to design a static, high-accuracy model, but to develop an AI multi-agent system that is economically sustainable, operationally efficient and capable of operating in a real-world production environment.

Accuracy This component of the score evaluates the system's capability to correctly detect fraudulent behaviour while minimizing unnecessary transaction blocks. The goal is to achieve a balanced trade-off between identifying fraud and avoiding unjustified patients impact.

The metric used is the algebraic mean between the $Value\ Recovery$ metric and the F_1 score:

$$Value\ Recovery = \frac{Correctly\ detected\ Fraud\ amount}{Original\ Fraud\ amount}$$

$$F_1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$$

with

$$Precision = \frac{TP}{TP + FP}, \qquad Recall = \frac{TP}{TP + FN}$$

where:

- TP: number of correctly detected frauds;
- FP: number of legitimate transactions incorrectly blocked;
- TN: number of legitimate transactions successfully processed;
- FN: number of undetected frauds.

Additional Metrics Cost, speed and efficiency serve as complementary evaluation metrics. They reward the design of an optimized agent architecture capable of detecting fraud in real time while maintaining low operational expense.

These dimensions emphasize the system's ability to perform under real-world constraints, balancing computational resources, processing latency and infrastructure usage. In essence, they measure how effectively your AI agents can scale, respond and adapt without compromising economic sustainability or responsiveness.

5 Constraints

- $T > 0$
- $T \in \mathbb{R}$